

AU25 Physics 5300 Final Project Proposal

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My final project is aimed to analyze and visualize a 3D simple cubic atomic lattice structure with 10-by-10-by-10 identical atoms($N = 10^3$) connected together with identical springs in all x, y, and z directions. I wish to analyze the normal modes in 3D structure and perturb part of the atoms to see how the perturbation wave propagates through the lattice. I could try to analyze the reflection of the wave from the boundary and more realistic concepts under it. The symmetry and conservation laws will also be explored in my system of atom lattices.

Python is the main computational tool to use for this project. Differential equation will be numerically solved. Related results will be interpreted both by plots and animations to ensure a good visualization. I will try to submit this project onto github.

To ensure a correct treatment of this problem, I will start with a smaller system(3-by-3-by-3) to at least check on the physical corrections of my derivation process and the function of my codes. After this, I will move into 10-by-10-by-10 system on analyzing the properties of such system. I hope there is a possibility for a short presentation at the end of the course.